

TrES-3b

R-2670

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110604 · created 2026-08-30T01:08:49+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455716.75984 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.018
Transit depth (Rp/Rs)^2	2.33 +/- 0.54 [%]
Orbital Inclination (inc)	83.64 +/- 0.8
Airmass coefficient 1 (a1)	1.201 +/- 0.017
Airmass coefficient 2 (a2)	-0.157 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.43 %
Best Comparison Star	None
Optimal Aperture	4.29
Optimal Annulus	14.41
Transit Duration (day)	0.0666 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.153 $\pm$ 0.018 vs 0.16309 (deviation 0.56 σ)
Duration fit vs published	0.0666 d vs 0.0567 d (deviation 1.95 σ)
Mid-time O-C	-3.0 min
Depth SNR	8.5
Residual scatter	1.43 %

Light curve

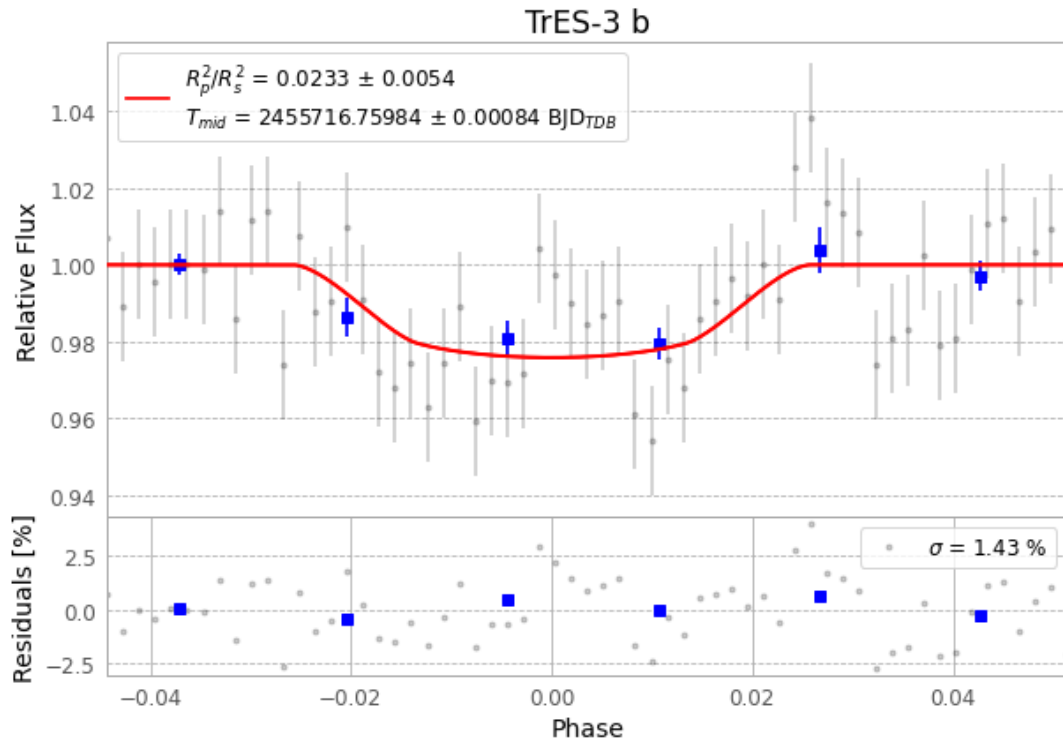


Figure 1 — FinalLightCurve_TrES-3 b_2011-06-03.png (SHA-256 7e98ddb41ec6a97d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 496b534ee92c6d88...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), TRES-3110604044513.FITS → TRES-3110604074513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110604.log (065a1f8bc943...), FinalLightCurve_TrES-3 b_2011-06-03.png (7e98ddb41ec6...), FinalLightCurve_TrES-3 b_2011-06-03.csv (a1d83f2fe24c...)

Compute resources

Wall time 112.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 01:08Z.